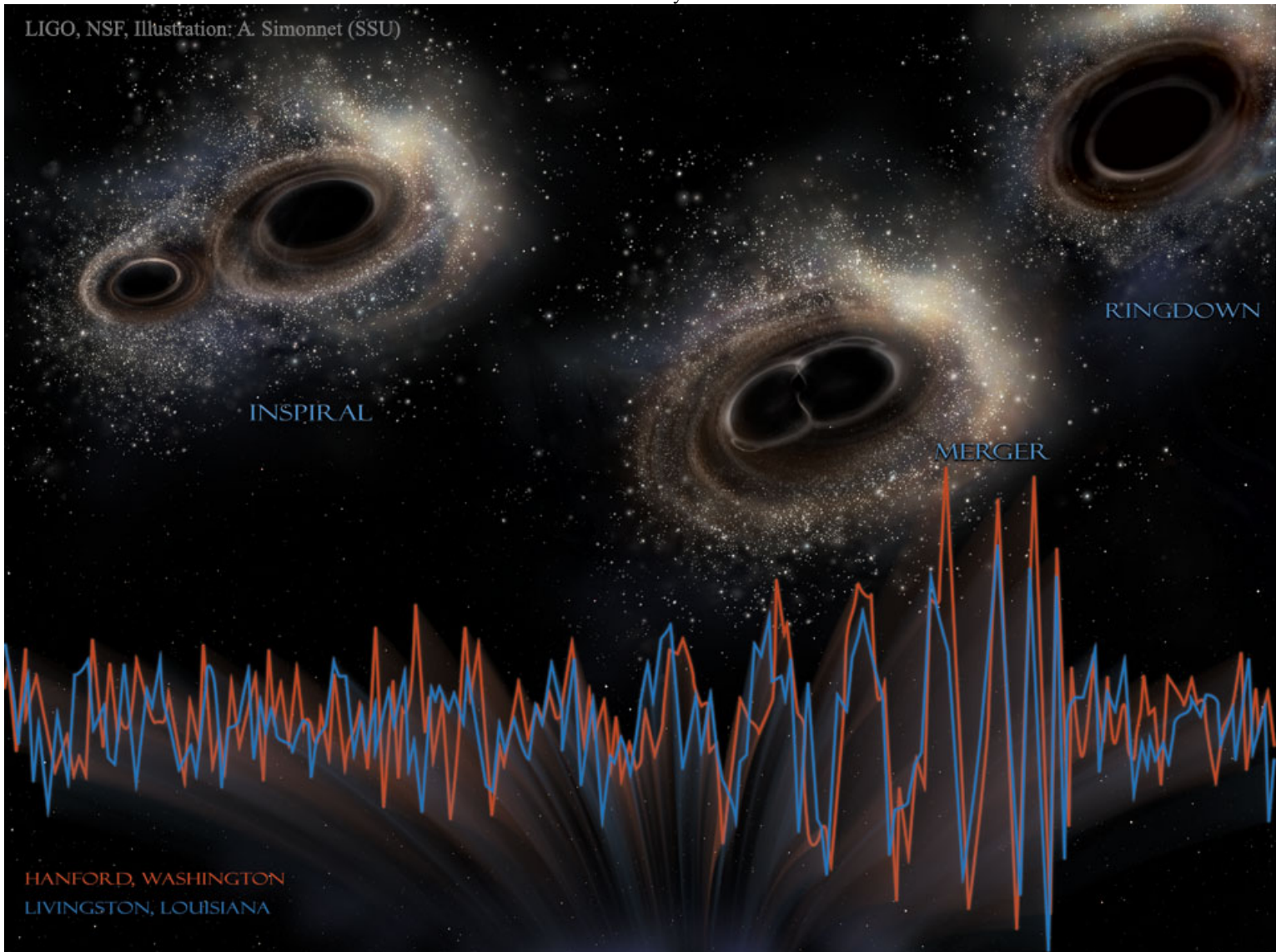


Astronomy Picture of the Day

[Discover the cosmos!](#) Each day a different image or photograph of our fascinating universe is featured, along with a brief explanation written by a professional astronomer.

2016 February 11



LIGO Detects Gravitational Waves from Merging Black Holes
Illustration Credit: [LIGO](#), [NSF](#), [Aurore Simonnet](#) ([Sonoma State U.](#))

Explanation: Gravitational radiation has been directly detected. The first-ever detection was made by both [facilities](#) of the [Laser Interferometer Gravitational-Wave Observatory](#) (LIGO) in [Washington](#) and [Louisiana](#) simultaneously last September. After numerous consistency checks, the resulting [5-sigma](#) discovery was [published today](#). The measured [gravitational waves](#) match those expected from two large black holes merging after a [death spiral](#) in a distant galaxy, with the resulting new black hole momentarily vibrating in a rapid [ringdown](#). A phenomenon [predicted](#) by [Einstein](#), the historic discovery confirms a cornerstone of humanity's understanding of [gravity](#) and basic physics. It is also the most direct detection of [black holes](#) ever. The featured illustration depicts the [two merging black holes](#) with the signal strength of the two detectors over 0.3 seconds superimposed across the bottom. Expected future detections by [Advanced LIGO](#) and [other gravitational wave detectors](#) may not only confirm the spectacular nature of this measurement but hold [tremendous promise](#) of giving humanity a [new way to see](#) and explore our universe.

Tomorrow's picture: [open spacetime](#)

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